



MADHU MARAJU

Chodavaram, Anakapalli, India

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Madhu Maraju – Engineering Portfolio

EDUCATION

Bachelor of Technology in Mechanical Engineering

2024 - 2027

Vishnu Institute of Technology (A), Bhimavaram | **CGPA: 9.09** (Gold Medalist)

Diploma in Mechanical Engineering

2021 - 2024

Government polytechnic, Tadepalligudem | **GPA: 9.4** (Gold Medalist)

Secondary School Education

2021

Government High School, Chodavaram | **GPA: 10**

INTERNSHIPS

INDIAN INSTITUTE OF TECHNOLOGY PALAKKAD (IIT PKD)

May-Jun 2026

Summer Research Intern (Stipend ₹12000/-) – Welding Technology Lab

Kerala, India

- Designed TPMS-based heat exchanger, regenerative rocket nozzle, and conformal cooling mold using Fusion 360 and nTop following DfAM principles.
- Implemented lattice structures and conformal cooling channels, and fabricated heat exchanger and rocket nozzle demonstration models through polymer-based 3D printing.

INDIAN INSTITUTE OF TECHNOLOGY BHUBANESWAR (IIT BBS)

Dec 2025

Winter Research Intern – School Of Mechanical Science (Sandy Group)

Bhubaneswar, India

- Developed a mathematical model of a vibro-impact system incorporating frictional effects for vibration control applications. Implemented non-dimensional governing equations in MATLAB and conducted numerical simulations. Analyzed system dynamics using time-history responses, phase portraits, and bifurcation diagrams.

INDIA YAMAHA MOTORS PVT.LTD.

June-July 2025

Summer Intern – Manufacturing Engineering (Export)

Chennai, India

- Contributed to CKD IPC digitalization and production process optimization, achieving ~10% cycle time reduction. And gained hands-on exposure to casting, machining, welding, painting, assembly, and basic CNC programming while supporting production operations.

SCHWING STETTER INDIA PVT.LTD.

Feb-Mar 2025

Academic Intern – Production Engineering

Chennai, India

- Analyzed concrete batching plant assembly workflows and identified process improvements with ~12% efficiency gain potential. Studied the working principles of load cells and proximity sensors, and gained practical exposure to industrial safety practices including 5S methodology.

THE CHODAVARAM CO-OPERATIVE SUGARS LTD

Nov 2023-May 2024

Diploma Trainee – Workshop

Anakapalli, India

- Assisted in sugar and co-generation plant operations and equipment maintenance, contributing to improved reliability of crushers, turbines, and boilers with reduced operational downtime.

PROJECTS

ASSET INTEGRITY FRAMEWORK FOR LIFE EXTENSION OF AGEING INDUSTRIAL ASSETS *(Team Lead of 5)*

- Developed a predictive maintenance framework using multi-parameter sensing, Health Score logic, and real-time monitoring dashboards, supported by ANSYS creep and fatigue analysis for condition-based inspection; secured Runner-Up (₹1,00,000) at the [ASME IMECE India 2025 Brain Bolt Event](#).

HEAVY LIFT SYSTEM DESIGN AND STRUCTURAL ANALYSIS | ASME India 2026 – Semi Finalist

- Designed a 45-tonne low-headroom lifting system using a fixed spreader bar and twin cantilever beams.
- Performed structural verification including CoG, sling, beam, and ballast calculations as per ASME lifting standards. And validated the design through CAD, ANSYS FEA, and lattice optimization.

PATIENT- SPECIFIC PROSTHETIC SOCKET USING 3D PRINTING

(Team Lead of 6)

- Selected among the **Top 15 out of 305 teams** in **SIH 2025** Internal Hackathon for developing a patient-specific prosthetic socket using Fusion 360, ANSYS, and FDM 3D printing based on MRI data.
- Fabricated a PLA based socket, reducing fabrication time by 80% and cost by 70% while enhancing comfort and strength.

SMART GLASS CLEANING ROBOT | IMECE India 2025

(Team Lead of 5)

- Designed a remote-controlled glass cleaning robot for high-rise buildings within 100 cm³ size constraints, integrating suction-based climbing, spray-roller cleaning and IoT-enabled wireless control and designed a lightweight carbon-fiber chassis using generative design.

ELECTROCHEMICAL ENERGY HARVESTING FROM SOLAR STILL

(Team Lead of 5)

- Designed and developed a hybrid solar still combining water purification and electricity generation. Enhanced solar still performance by integrating electrochemical cells, delivering 0.8–1.2 V per cycle and ~25% improvement in water output.

SKILLS

- CAD Tools:** AutoCAD | CATIA V5 | Fusion360 | SolidWorks | Rhino3D.
- Simulation & Analysis Tools:** ANSYS (structural analysis and CFD) | MATLAB | nTop.
- Presentation tools:** MS Word | MS Excel | MS PowerPoint | Canva.
- Soft Skills:** Technical Documentation | Adaptability | Team Collaboration.
- Languages:** English | Telugu (Native) | Tamil | Japanese (Beginner level) | Hindi.

CERTIFICATIONS

- Manufacturing Technology I & II (Elite) – [NPTEL](#), Operations Management (Elite) – [NPTEL](#)
- MATLAB** onramp completion certificate – [MATLAB](#).
- 3D Printing** Fundamentals – [HP LIFE](#).
- [CFD & Heat Transfer 3 - Day Workshop](#) – BVRIT Narsapur, Conducted by IIT JAMMU Experts.

ACHIEVEMENTS

- Runner-Up (2nd Prize), ASME IMECE India 2025** – Brain Bolt Event, Hyderabad, India | 10th – 12th September 2025.
- Finalist** – Space Technologies for Earth & Beyond 2025: Viksit Bharat Ideathon ([IIT Tirupati](#)) – jointly organized by ISRO–NRSC, Tirupati, India.
- Top 10 Finalist**, Regional-Level Techfest 2022, Vijayawada, India | November 2022 – Qualified for the [National-Level Competition](#).
- Secured Second Prize (₹2,000), **Researcher's Day 2026** – Poster Presentation on Nonlinear Vibration Control (VINES), Vishnu Institute of Technology, Bhimavaram, India | 2026.
- Secured Second Prize (₹2,000), **InnovIT-2K26 Project Expo**, Vishnu Institute of Technology, Bhimavaram, India | 2026.